RELAY

Incident triage for logistics

BUSINESS SUMMARY / INDEPENDENT SYNTHETIC TEST / 21 OCTOBER 2026

The decision

Keep routing **human-confirmed**. The preregistered Rules / Sensitive candidate missed two urgent cases. Text / Sensitive had the lowest observed test cost, but selecting it now requires confirmation on a new set before operational use.

Problem and working prototype

Short incident messages mix delivery, missing cargo, damage, cold-chain, safety and document issues. A missed urgent flag can delay a critical response. Relay proposes a receiving team and urgency, flags uncertainty, and records human corrections. Urgent review cases stay in Urgent; ordinary Needs review is separate. No real dispatch or alert is sent.

Evidence

Two approaches: an explicit rules baseline and a trained TF-IDF text classifier. Training uses 240 reports in 48 scenario families. A separate agent reviewed 60 held-out scenarios using only text and the guide. Its judgments were frozen before scoring; these are not domain-expert human labels.

Approach / policy	Macro F1	Urgent recall	Review	Cost*	p95 ms
Rules / Standard	0.551	6%	27/60	1601	0.105
Rules / Sensitive	0.551	88%	54/60	310	0.101
Text / Standard	0.648	62%	19/60	717	0.024
Text / Sensitive	0.648	100%	42/60	76	0.024

*Relative cost with miss penalty 100 for every row. Category and urgency each have 56 known labels; urgent recall uses 16 urgent cases. Four unknowns in each field remain unknown. Review uses all 60 reports. p95 excludes HTTP, startup and human decisions.

Changed requirement and workload

Raising the missed-urgent penalty from 30 to 100 favors stricter review. Rules changes from 15 to 2 misses and 27 to 54 reviews; cost falls from 1601 to 310 at the same new penalty. Text / Sensitive catches 16/16 urgent cases, but flags 8 routine cases urgent and reviews 42/60. Its two wrong eligible routes and two bypassed required reviews remain material failures.

Economics, limits and next step

Both classifiers incur **USD 0 in inference API calls**; compute, development and review service/labor costs are unmeasured and excluded. Relative assumptions: review 1, wrong eligible route or bypassed required review 6, false urgency 2, missed priority 30 or 100 even during ordinary review. These are scenario assumptions, not measured savings. Small synthetic data, shared AI tooling, guide ambiguity and unmeasured review delays limit the evidence.

Next: obtain representative operator reports and domain review, test the next candidate on a new set, and measure review capacity and escalation outcomes. No field trial has occurred.

Evidence: part-b/evidence/final-test.json; DATA.md; EVALUATION.md. Product direction and prototype feedback: dansyar. Software, synthetic data and documents: OpenAI Codex. Independent labels: separate fresh-context Codex agent. Full attribution: ATTRIBUTION.md.